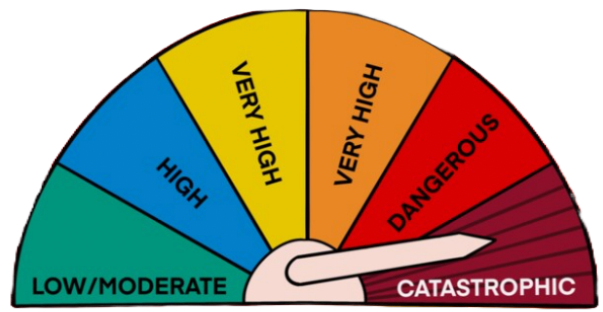
Highly Accurate Expectation Values using High-Order Relativistic Coupled Cluster Theory

Gabriele Fabbro

CNRS/Université Toulouse - Paul Sabatier

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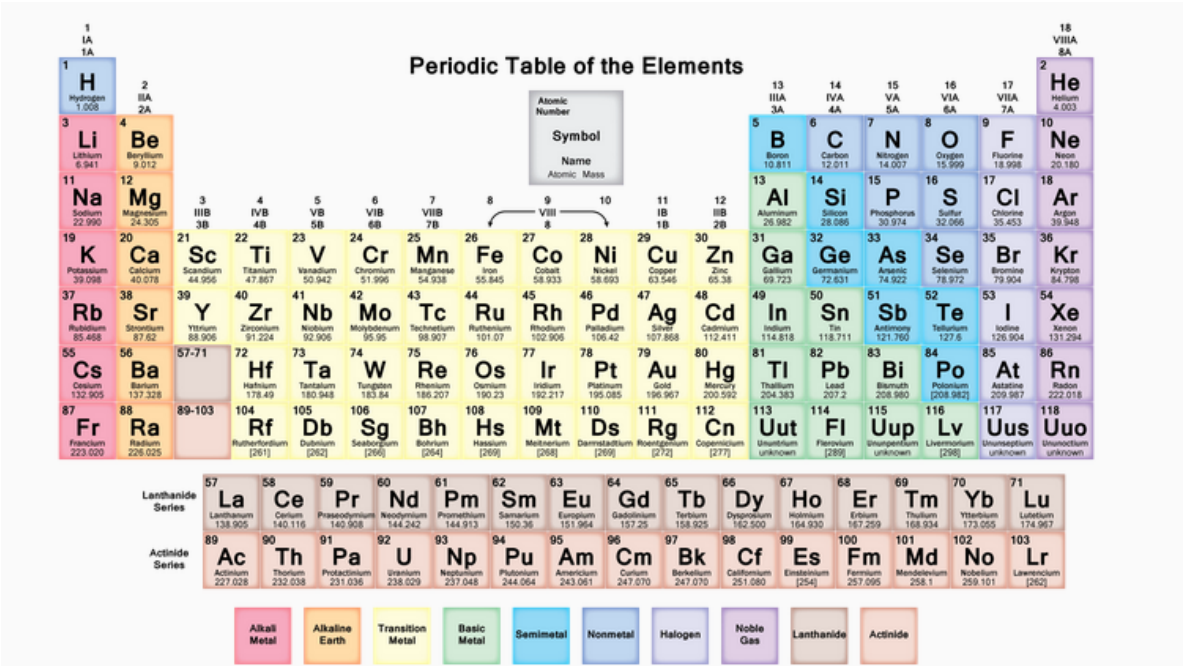
June 11, 2025



Today's CC level

Introduction

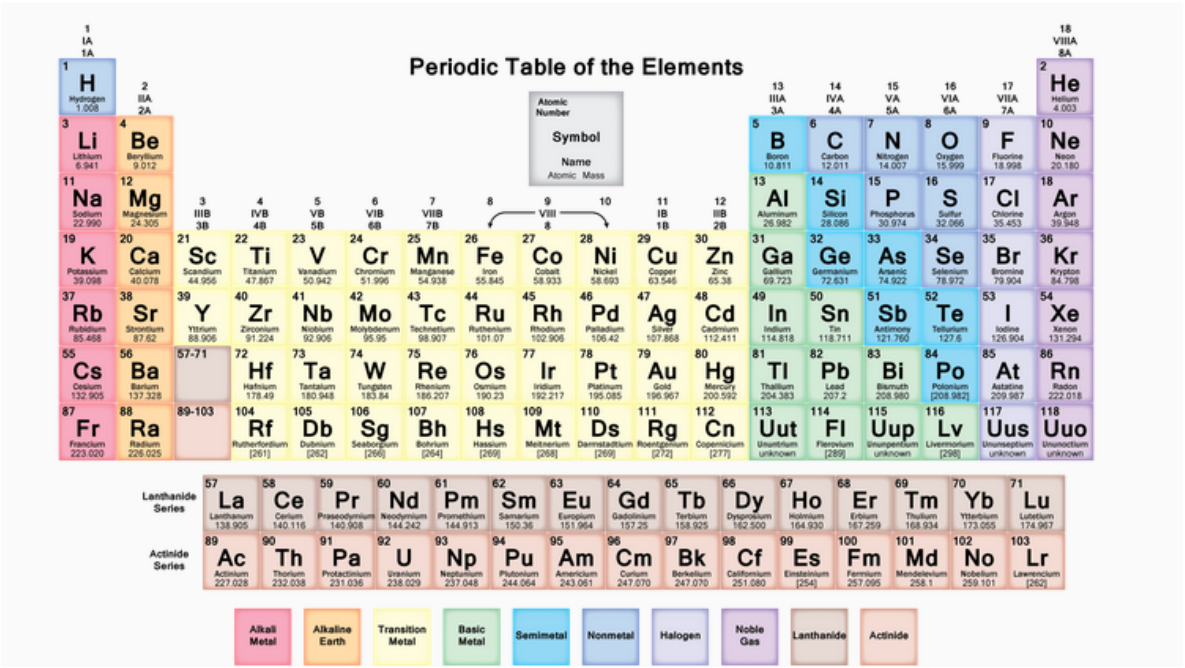
Our playground



- Elements of the periodic table are described using quantum mechanics.

Introduction

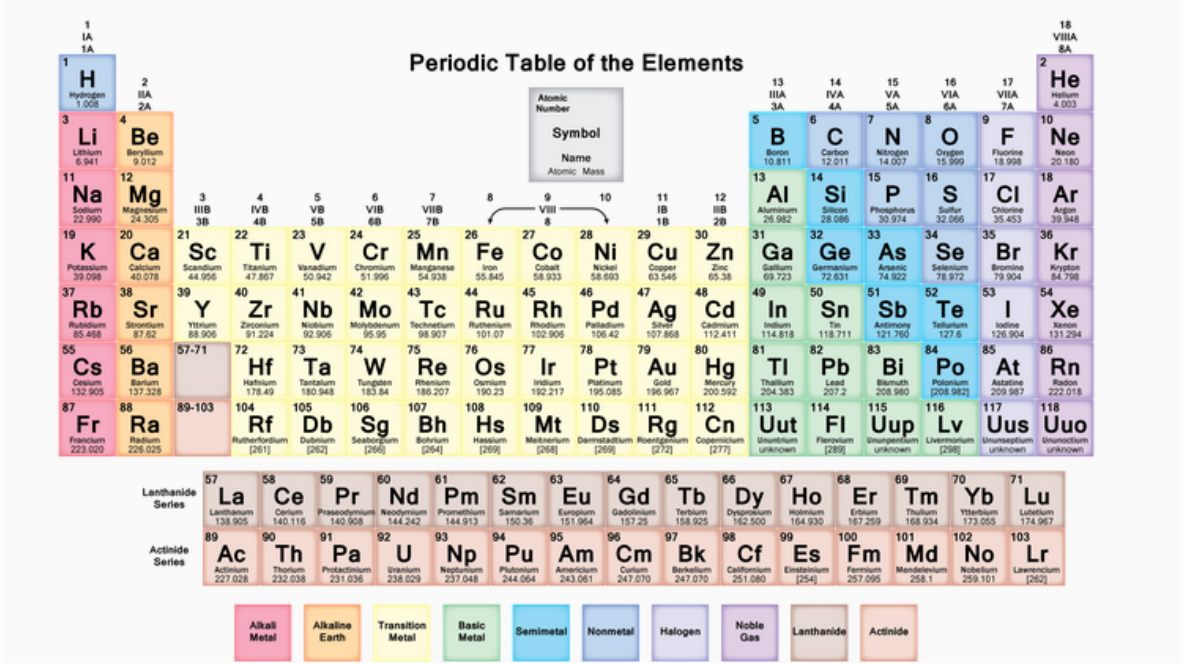
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- Elements of the periodic table are described using quantum mechanics.
- Heavy elements need *special* considerations:

Introduction

Our playground



- Elements of the periodic table are described using quantum mechanics.
- Heavy elements need *special* considerations:

HAMP-vQED project

– **Correlation effects**
– **Relativistic effects**
– **QED effects**

Coupled Cluster Theory

A non-complete introduction

Exponential operator

$$|\Psi_{\mathrm{CC}}\rangle = e^{\hat{T}} |\Phi_0\rangle \tag{1}$$

CC wave function

Reference

Coupled Cluster Theory

A non-complete introduction

Exponential operator

$$|\Psi_{\mathrm{CC}}\rangle = e^{\hat{T}} |\Phi_0\rangle \quad , \quad \hat{T} = \sum_{\mu} t_{\mu;k} \quad \hat{\tau}_{\mu;k} \tag{1}$$

CC wave function — Reference — Amplitudes — Excitation operators

Coupled Cluster Theory

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Excited Determinant

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CC wave function — Reference — Amplitudes — Excitation operators

Normal-ordered Hamiltonian

$$\hat{H}_{\mathrm{N}} = \hat{H} - \langle\Phi_0|\hat{H}|\Phi_0\rangle = \sum_{pq} f_{pq}\{a_p^\dagger a_q\} + \frac{1}{4}\sum_{pqrs} V_{pq}^{rs}\{a_p^\dagger a_q^\dagger a_s a_r\} \tag{2}$$

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similarity-transformed Hamiltonian:

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Projecting

$$\langle\Phi_0| \quad : \quad \langle\Phi_0|\hat{\overline{H}}|\Phi_0\rangle = E_{\mathrm{CC}}(\{\boldsymbol{t}\}) \leftarrow \text{Energy (not an upper bound)} \tag{4}$$

$$\langle\Phi_{\mu;k}| \quad : \quad \langle\Phi_{\mu;k}|\hat{\overline{H}}|\Phi_0\rangle = 0, \quad k \leq k_{\max} \leftarrow \text{Amplitudes} \tag{5}$$

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Approximations

$$k_{\max} = 2 \rightarrow \mathrm{CCSD}, \quad k_{\max} = 3 \rightarrow \mathrm{CCSDT}, \ldots \tag{6}$$

CC expectation values

For approximate wave-functions, expectation values are computed by

$$\hat{H}(\boldsymbol{\varepsilon}) = \hat{H}_0 + \hat{V}(\boldsymbol{\varepsilon}) \rightarrow \left.\frac{dE}{d\boldsymbol{\varepsilon}}\right|_{\boldsymbol{\varepsilon}=0} = \left[\frac{\partial E}{\partial \boldsymbol{\varepsilon}} + \frac{\partial E}{\partial \boldsymbol{\lambda}}\frac{d\boldsymbol{\lambda}}{d\boldsymbol{\varepsilon}}\right]_{\boldsymbol{\varepsilon}=0} \tag{7}$$

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CC expectation values

For CC wave-functions, we get

$$\left.\frac{dE_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=0} = \left[\frac{\partial E_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{\partial \varepsilon_A} + \frac{\partial E_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{\partial t_\nu}\frac{dt_\nu}{d\varepsilon_A}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{7}$$

CC expectation values

...but

$$\left.\frac{dE_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=0} = \left[\frac{\partial E_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{\partial \varepsilon_A} + \underbrace{\frac{\partial E_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{\partial t_\nu}}_{\neq 0}\frac{dt_\nu}{d\varepsilon_A}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{7}$$



Figure 1: KC Green's web comic strip "On Fire"

CC expectation values

We are using the Lagrangian framework:[1][2]

$$\left.\begin{array}{l}\langle\Phi_0|\,\hat{\overline{H}}\,|\Phi_0\rangle = E_{\mathrm{CC}} \\ \langle\Phi_\mu|\,\hat{\overline{H}}\,|\Phi_0\rangle = 0\end{array}\right\} \rightarrow L^{\mathrm{CC}} = \langle\Phi_0|\,(1+\hat{\Lambda})\hat{\overline{H}}\,|\Phi_0\rangle\,; \quad \hat{\Lambda} = \sum_\nu \bar{t}_\nu \hat{\tau}_\nu^\dagger \qquad (8)$$

[1] T. Helgaker and P. Jørgensen, *Theor. Chim. Acta*, **75**, 111-127 (1989)

[2] H. Koch and co-workers J. Chem. Phys. 92 (1990) 4924

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Variational conditions

$$\frac{\partial L^{\rm CC}}{\partial \bar{t}_\mu} = \langle\Phi_\mu|\,\hat{\overline{H}}\,|\Phi_0\rangle = 0 \rightarrow \text{Amplitude Eqs.} \tag{9}$$

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Expectation values

$$\hat{H} = \hat{H}_0 + \sum_X \varepsilon_X \hat{H}_X \rightarrow \left.\frac{dL^{\rm CC}}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=0} = \left.\frac{\partial L^{\rm CC}}{\partial \varepsilon_A}\right|_{\boldsymbol{\varepsilon}=0} = \sum_{pq} h_{pq;A}\gamma_{pq}^{\rm CC} \tag{11}$$

$$\gamma_{pq}^{\rm CC} = \langle\Phi_0|(1+\hat{\Lambda})\overline{\{a_p^\dagger a_q\}}|\Phi_0\rangle \rightarrow \text{One-body CC density matrix} \tag{12}$$

---

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Generation of the CC equations

Several ways:

- Second quantization
- Wick's theorem
- **Diagrams**

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CCSDTQP	$\mathcal{O}(N^{12})$	289

Table 1: CC amplitude equations: scaling and number of diagrams

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Problems with High-Order excitations

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1. Cumbersome equations;
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$$R_i^a = \sum_{jk,bc} A_{bc}^{jk} B_j^b C_{ik}^{ac}, \quad \mathcal{O}(o^3v^3)$$

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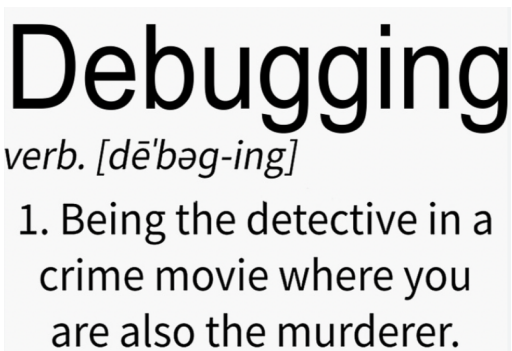
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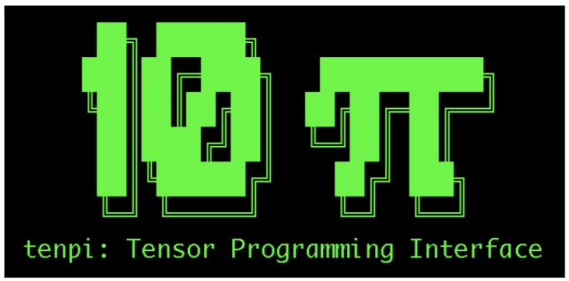
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3. Debugging challenges.



tenpi: elegant development of CC methods

J. Brandejs, J. Pototschnig and T. Saue, (2024). ***Generating coupled cluster code for modern distributed memory tensor software.*** `arXiv:2409.06759`. ***J. Chem. Phys. A*, accepted.**



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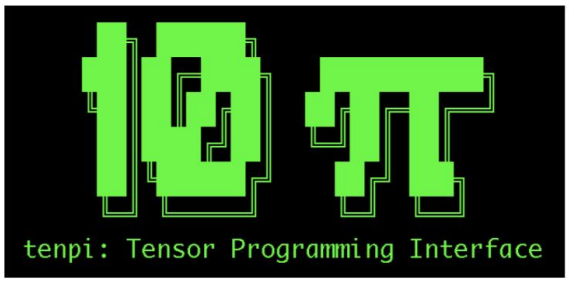


- General-order CC generator.
- Automatic optimization of intermediates.
- Export compatible code (Fortran,...) with calls to tensor libraries (TALSH,..).
- Well integrated within the DIRAC program package.

T. Saue et al., *J. Chem. Phys.*, **152**, (2020).

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Functionalities

1. General-order CC;
2. Perturbative T,Q;
3. Matrix elements;
4. **Properties**

tenpi: elegant development of CC methods

Equation generator principle

- Based on the diagrammatic scheme of M. Kállay and P. R. Surján
  - M. Kállay and P. R. Surján, *J. Chem. Phys.*, **115**, 2945 (2001).
- Integer strings of length 13 are used to represent a diagram.

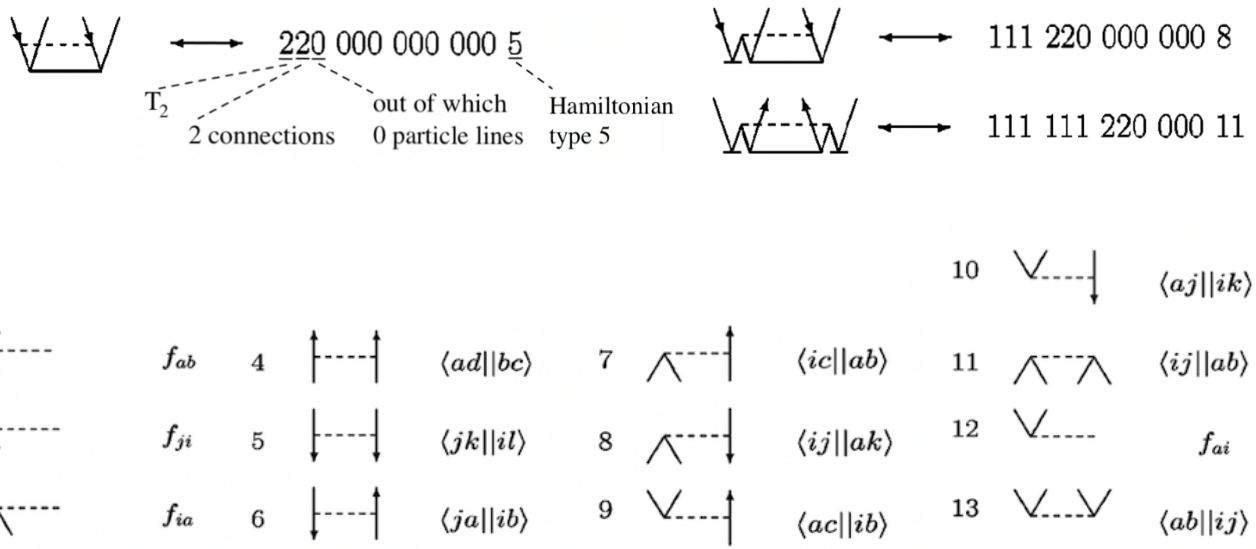
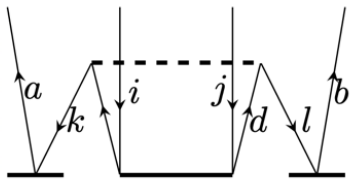


Figure 2: From M. Kállay and P. R. Surján, *J. Chem. Phys.*, **115**, 2945 (2001)

tenpi: elegant development of CC methods

workflow

Generate only unique diagrams



Translate to line graph representation

```
diag.lines.append([node1, node2])
```

Plot diagrams PDF

Apply rules to obtain equation terms

$$\frac{1}{4}\, P(ab)\, V_{cd}^{kl}\, t_k^a\, t_l^b\, t_{ij}^{cd}$$

Plot PDF

Apply permutations

Devise intermediates (op. count)

$$D_1{}_i^k = f_c^k\, t_i^c$$

$$r_{ji}^{ba} = t_{jk}^{ba}\, D_1{}_i^k$$

'Remove' selected intermediates

Test correctness: brute-force contr.

```
for h1 in range (0, 0):
 for p5 in range (0, V):
  temp_4[h4][h5][h2][h1] +=
  t_vo[p5][h2] * temp_3[h4][h5][p5][h1]
```

Generate code: instructions for tensor lib.

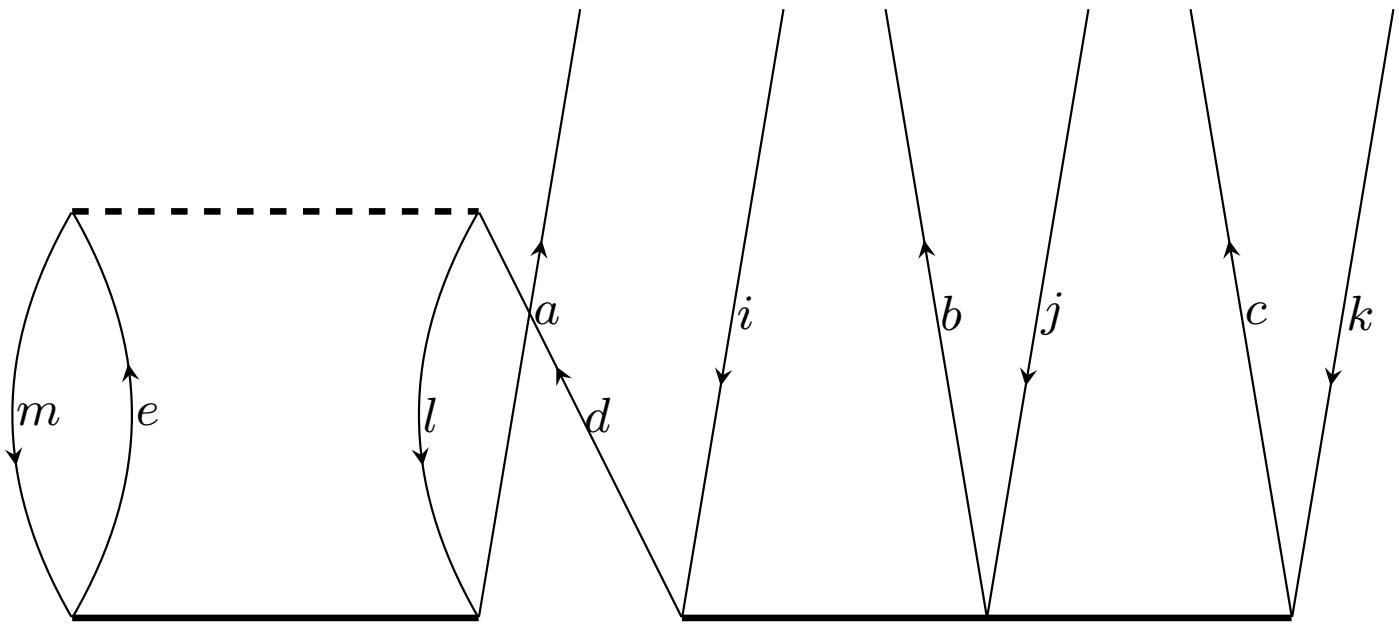
```
talsh_tensor_contract(R2,T2,D1,
"R2(b1,a1,j1,i1)+=T2(b1,a1,j1,k1)*D1(k1,i1)")
```

Diagrams for CC expectation values

M. Kállay, J. Gauss, and P. G. Szalay, J. Chem. Phys. 119 (2003) 2991

J. Gauss and J. F. Stanton, J. Chem. Phys. 116 (2002) 1773

- From $\langle \Phi_{ijk}^{abc} | \hat{\overline{H}} | \Phi_0 \rangle = 0$ we take, as an example, the following diagram



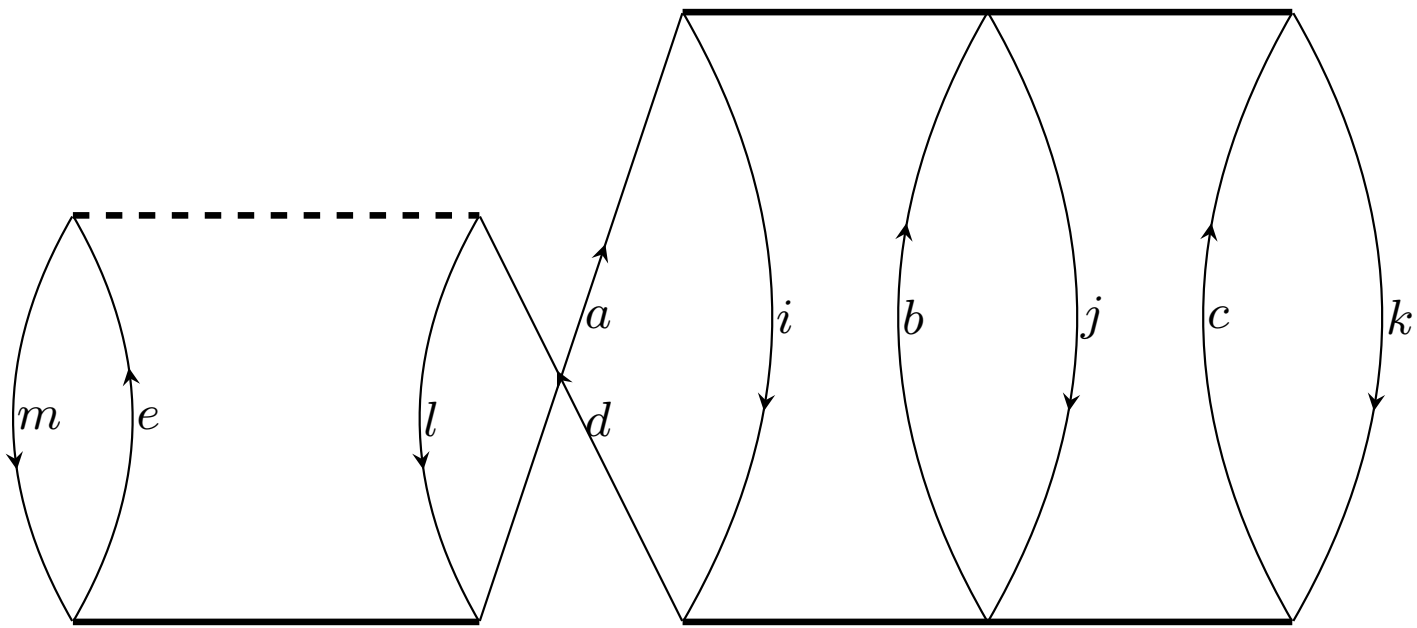
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- We introduce the CC Lagrangian:

$$L^{\mathrm{CC}} = E^{\mathrm{CC}} + \sum_{\mu} \bar{t}_{\mu} \langle \Phi_{\mu} | \hat{\overline{H}} | \Phi_0 \rangle = \langle \Phi_0 | (1 + \hat{\Lambda}) \hat{\overline{H}} | \Phi_0 \rangle \,, \qquad \hat{\Lambda} = \sum_{\mu} \bar{t}_{\mu} \hat{\tau}_{\mu}^{\dagger}$$

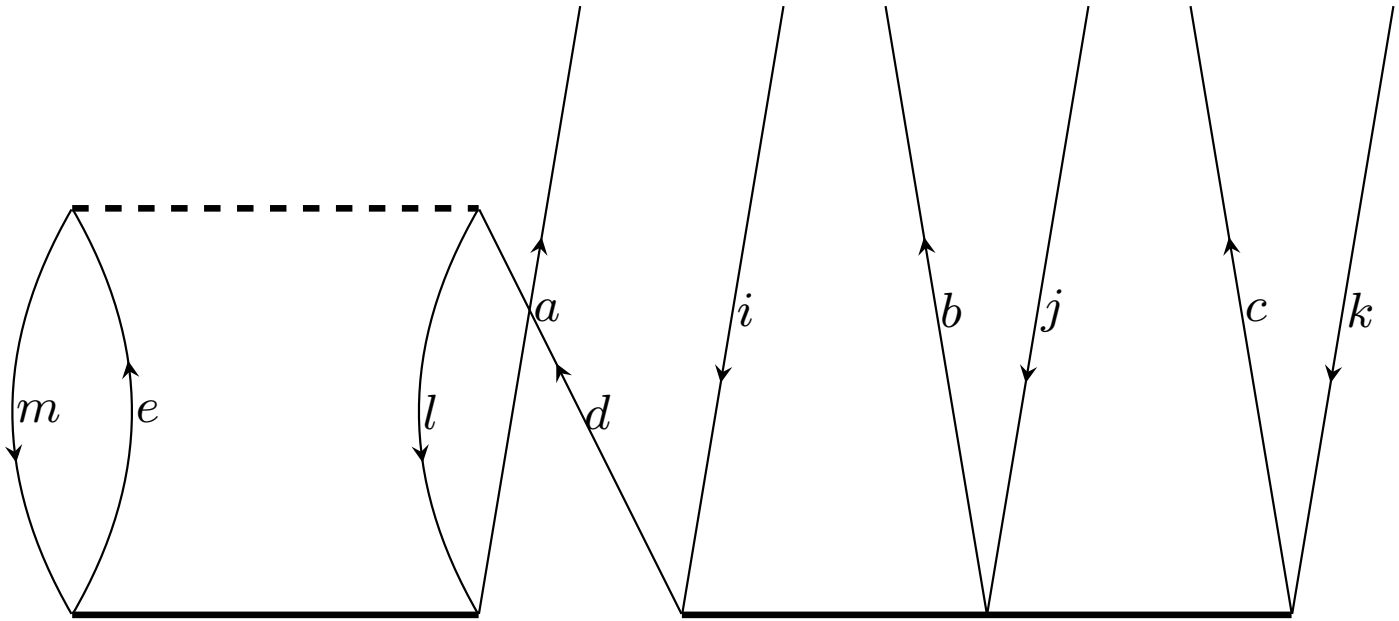


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- Amplitude equations: $\frac{\partial L^{\mathrm{CC}}}{\partial \bar{t}_\mu} = \langle \Phi_\mu | \hat{\overline{H}} | \Phi_0 \rangle = 0.$

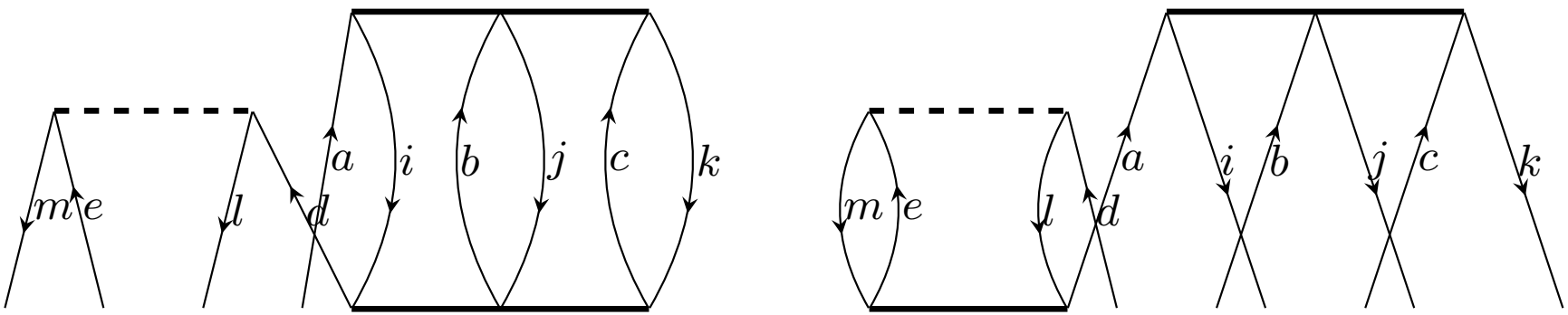


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- $\Lambda-$equations: $\frac{\partial L^{\mathrm{CC}}}{\partial t_\mu} = \langle \Phi_0 | (1 + \hat{\Lambda}) [\hat{\overline{H}}, \hat{\tau}_\mu] | \Phi_0 \rangle = 0.$



Diagrams for CC expectation values

G. Fabbro, J. Brandejs and T. Saue, arXiv:2504.18516

- Not everything is connected...

$$\frac{\partial L_{\mathrm{CC}}}{\partial t_\nu} = \langle \Phi_0 | (1 + \hat{\Lambda}) [\hat{\overline{H}}, \hat{\tau}_\nu] | \Phi_0 \rangle = \langle \Phi_0 | \hat{\overline{H}} | \Phi_\nu \rangle + \langle \Phi_0 | [\hat{\Lambda}, \hat{\overline{H}}] | \Phi_\nu \rangle + \sum_\eta \underbrace{\langle \Phi_0 | \hat{\overline{H}} | \Phi_\eta \rangle \langle \Phi_\eta | \hat{\Lambda} | \Phi_\nu \rangle}_{\text{Disconnected terms}} \tag{13}$$

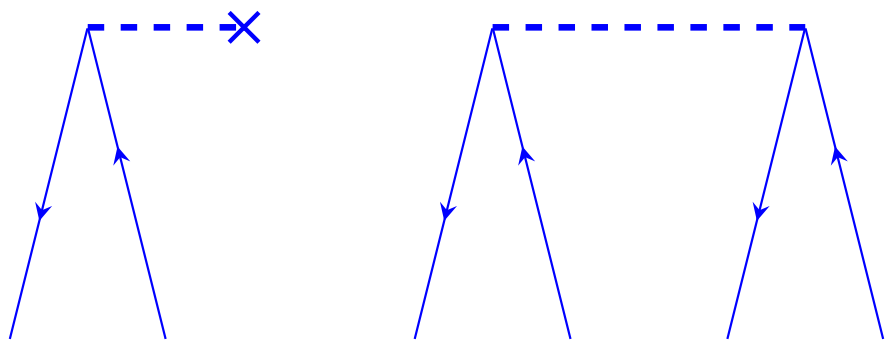
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- Not everything is connected...

$$\frac{\partial L_{\mathrm{CC}}}{\partial t_\nu} = \langle \Phi_0 | (1 + \hat{\Lambda})[\hat{\overline{H}}, \hat{\tau}_\nu] | \Phi_0 \rangle = \langle \Phi_0 | \hat{\overline{H}} | \Phi_\nu \rangle + \langle \Phi_0 | [\hat{\Lambda}, \hat{\overline{H}}] | \Phi_\nu \rangle + \sum_\eta \underbrace{\langle \Phi_0 | \hat{\overline{H}} | \Phi_\eta \rangle \langle \Phi_\eta | \hat{\Lambda} | \Phi_\nu \rangle}_{\text{Disconnected terms}} \tag{13}$$

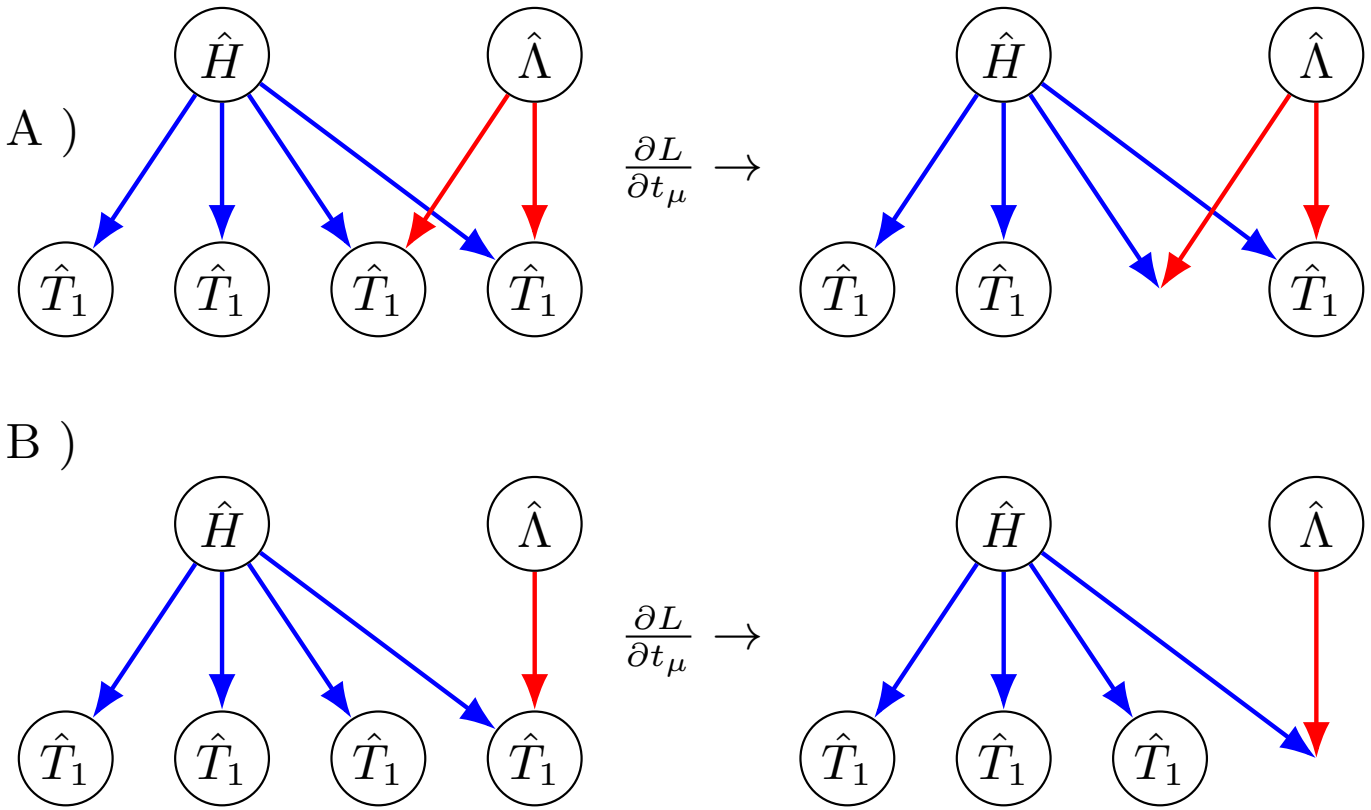
- Disconnected diagrams can only occur when the $\hat{\Lambda}$-operator is not directly connected to the Hamiltonian, which limits the possible elements to:



Diagrams for CC expectation values

G. Fabbro, J. Brandejs and T. Saue, arXiv:2504.18516

- In order to be disconnected upon removal of a $\hat{T}$-interaction line, the $\hat{\Lambda}$ has to be connected to that $\hat{T}$ only, hence by an *even* number of connections, while the $\hat{T}$ at the same time must be connected to the Hamiltonian.

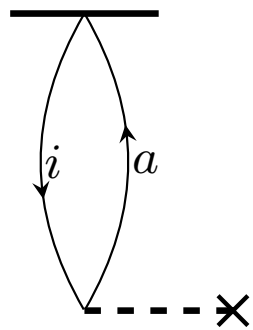


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M. Kállay, J. Gauss, and P. G. Szalay, J. Chem. Phys. 119 (2003) 2991

J. Gauss and J. F. Stanton, J. Chem. Phys. 116 (2002) 1773

- One-body CC density matrices: $\gamma_{pq}^{\mathrm{CC}} = \langle \Phi_0 | (1 + \hat{\Lambda}) \overline{\{a_p^\dagger a_q\}} | \Phi_0 \rangle$
- The topological part is equal to $\hat{F}_N = \sum_{pq} f_{pq} \{a_p^\dagger a_q\}$



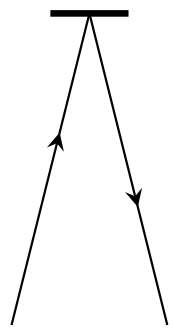
$$\langle \Phi_0 | \hat{\Lambda}_1 \hat{F}_N | \Phi_0 \rangle = \sum_{ia} \bar{t}_a^i f_i^a$$

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- One-body CC density matrices: $\gamma_{pq}^{\mathrm{CC}} = \langle \Phi_0 | (1 + \hat{\Lambda}) \overline{\{a_p^\dagger a_q\}} | \Phi_0 \rangle$
- The second-quantized part is equal to $\hat{F}_N = \sum_{pq} f_{pq} \{a_p^\dagger a_q\}$



$$\langle \Phi_0 | \hat{\Lambda}_1 \{a_p^\dagger a_q\} | \Phi_0 \rangle = \bar{t}_a^i \equiv \gamma_{ai}^{\mathrm{CC}}$$

NQMs from molecular calculations

G. Fabbro, J. Pototschnig and T. Saue, *J. Phys. Chem. A* (2025).

H. Guan et al., arXiv:2403.06384 [physics.atom-ph] (2024), accepted in *Phys. Rev. A Letter*.

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What we learned from the Electric Field Gradient (EFG)

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We extracted the NQMs of ^{7}Li and ^{27}Al

- $Q(^{7}\mathrm{Li}) = 0.0386$ b, in excellent agreement with the recent determination by Guan and co-workers.
- $Q(^{27}\mathrm{Al}) = 0.1466$ b in good agreement with previous findings by Pernpointner and Visscher.

Conclusions and Perspectives

What we have done

- Generation of arbitrary high-order CC expectation values using `tenpi`.
- First implementation of CCSDT and CCSDTQ expectation values in `DIRAC`;

  → **Check out the full paper for more and applications**



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Perspectives

- Multiple-node implementation;
- Point-group symmetry;
- Relaxation effects;

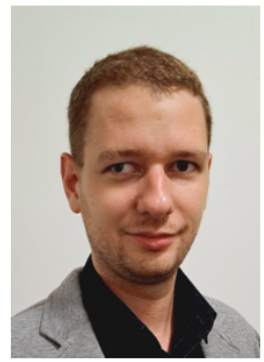
HAMP-vQED team



Trond Saue








Jan Brandejs, Johann Pototschnig, Gabriele Fabbro (CC team)

Maen Salman, Dávid Ferenc, Ryan Benazzouk (QED team)

Q(^{7}Li) from relativistic molecular calculations

G. Fabbro, J. Brandejs and T. Saue, arXiv:2504.18516

	LiH		LiF		LiCl	
	EFG	Q	EFG	Q	EFG	Q
HF	-0.0397	-0.0371	-0.0490	-0.0361	-0.0275	-0.0387
CCSD	-0.0387	-0.0382	-0.0460	-0.0384	-0.0275	-0.0387
+ T	0.0010	0.0015	0.0003	-0.0002	0.0003	-0.0004
+ Q	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	-0.0377	-0.0367	-0.0457	-0.0386	-0.0272	-0.0391

Table 2: Individual contributions to the EFG (E_h/ea_0^2) at the Li position for the systems LiX (X = H, F, Cl) and the derived values for the ^{7}Li quadrupole moment (Q, in mb).

- By adding the rovibrational contribution (about 2–3% in Q) and performing a best fit Q(^{7}Li)=-0.0386 b.
- Perfect agreement with recent experiments.[3]

[3] H. Guan et al., arXiv:2403.06384 [physics.atom-ph] (2024)

Q(^{27}Al) from relativistic molecular calculations

G. Fabbro, J. Brandejs and T. Saue, arXiv:2504.18516

	AlH		AlF		AlCl		AlBr	
	EFG	Q	EFG	Q	EFG	Q	EFG	Q
HF	-1.4722	0.1405	-1.1926	0.1347	-0.9819	0.1318	-0.9070	0.1314
CCSD	-1.4209	0.1455	-1.1066	0.1452	-0.8977	0.1442	-0.8282	0.1439
+ T	0.0086	0.0012	0.0087	0.0012	0.0065	0.0010	0.0058	0.0010
+ Q	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	-1.4119	0.1467	-1.0979	0.1464	-0.8912	0.1452	-0.8224	0.1449

Table 3: Individual contributions to the EFG (in E_h/ea_0^2) at the Al position for the systems AlY (Y = H, F, Cl, Br) and the derived values for the ^{27}Al quadrupole moment (Q, in mb).

- By adding the rovibrational contribution (about 0.5-2% in Q) and performing a best fit Q(^{7}Li)=0.1466 b.
- Supported by results obtained by Pernpointner and Visscher.[4]
- Triple excitations still influence Q with a change of 0.7-0.8%.

[4] M. Pernpointner and L. Visscher, J. Chem. Phys. 114, 10389–10395 (2001)